



# EcoCampus

**AI-powered Smart & Green Campus Platform**

**Building a Closed-Loop Circular Economy for Sustainable Campuses**



**Team 1 – GCNB**

**Green · Code · No Bug**

**University of Computer Studies, Mandalay (UCSM)**

# THE PROBLEM

When graduation arrives or semesters end, why do tons of fully functional items end up in campus trash cans instead of find new owner.



## Resource Waste

Books, clothing, and electronics are often discarded into landfills.



## Disposal Confusion

Students lack clear guidance on reusing, reusing, recycling, or recycling, or donating items.



## Lack of Motivation

Eco-friendly actions lack visible incentives and feedback.

VALUABLE RESOURCES → WASTED POTENTIAL



# OUR UNIQUE INSIGHT

Sorting waste is only half the battle—the true key is an incentive-driven circular economy.

## Beyond Trash Sorting



From disposal to lifecycle care

## Value-Driven Habit



Green actions = real value

## Hyper-Local Ecosystem



Trusted student-only network.

We don't just sort waste — we unlock value.





# OUR INNOVATION

Not Just Another Second-Hand Market:  
The Green Points Advantage.





# THE SMART ENGINE: AI INTEGRATION

## AI Integration Beyond a Basic Chatbot

### Visual Object Recognition

Instantly identifies uploaded items via computer vision.



Instantly identifies uploaded items via computer vision.

### Smart Eco-Routing

AI analyzes item condition and recommends the optimal disposal path, Reuse, Recycle, or Donate.



### Dynamic Valuation



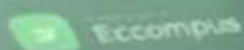
Suggests fair Green Point values based on item condition and category.

### Intelligent Demand Matching

Connects idle resources with students who actively need them.



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# EcoCampus Ecosystem

One Campus. One Green Economy.

EcoCampus connects environmental actions with a gamified campus marketplace. Together, we create value, community, and impact.



1



## Campus Registration

Students sign up with Student ID, Email, Password, and Year / Batch.

New users receive Initial Exchange Coins.

2



## Green Points

Earn Green Points through everyday environmental actions.

- Recycling
- Waste Sorting
- Green Activities
- Volunteering
- AI Recycling Scan

Green Points represent your environmental contribution.

3



## Year / Batch Leaderboard

Green Points are accumulated by Year / Batch and displayed on a ranked leaderboard.

Competition Period  
→ Ranking →  
Friendly Competition

4



## Exchange Coins

Convert Green Points into Exchange Coins.



Exchange Coins are used in the Marketplace.

Converting Green Points into Exchange Coins does not affect your Leaderboard score.

5



## Reuse Marketplace

Buy, sell, or trade used items with Exchange Coins.

- Textbooks
- Stationery
- Clothing
- Daily-use Items
- ...and more

## THE ECOCAMPUS LOOP



**ACT**  
Take green actions.

**EARN**  
Earn Green Points.

**EXCHANGE**  
Convert to Exchange Coins.

**REUSE**  
Exchange or reuse items.

**IMPACT**  
Create a greener campus.

ACT → EARN → EXCHANGE → REUSE →  
IMPACT



# Smart & Connected Campus

Where AI Meets the Green Campus

EcoCampus integrates AI-powered tools, community connections, and real-time data to build a smarter, more sustainable campus.



## AI Recycling Classification

Point, shoot, and upload an image of a recyclable item. Gemini AI automatically classifies the item and awards Green Points instantly.



## Ask AI

Ask anything about the environment, campus life, or projects. Gemini AI is here to help you.

Environmental Questions

Campus Questions

Project / General Questions

Ask → Understand → Take Better Action

## Campus & Yearly Chat Feeds

Connect and communicate within your campus community.

### Year / Batch Feed

Chat exclusively with students from the same Year / Batch.

### Campus Feed

A general feed for all campus users to share and engage.

Share ideas · Discuss · Organize · Stay Connected

## Admin Dashboard

Admin can monitor and manage the platform with a comprehensive dashboard.

 Registered Users

 Listed Items

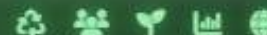
 Recycling Images

Real-time data · Smart insights · Better decisions



 Powered by Gemini (gemini-3.6-flash)

 Building a greener campus through technology, community, and continuous impact.



# TRADITIONAL MARKETPLACE

Trade with Money. Value is limited.



## TRADITIONAL SECOND-HAND MARKETPLACE



**Transaction Medium**  
Real Money



**Value Source**  
Money



**Purpose**  
Buy / Sell to save or make money



**User Behavior**  
Buy → Sell



**Incentive Mechanism**  
No environmental incentive



**Environmental Impact**  
Limited impact



**AI Assistance**  
Usually not a core function



**Resource Flow**  
Second-hand circulation only



**Core Value**  
Economic Value

### TRADITIONAL FLOW



Have an item



Sell it



Receive money



Buy another item



# ECO CAMPUS GREEN POINT MARKETPLACE

Trade with Green Actions. Value is unlimited.



## ECOCAMPUS GREEN POINT MARKETPLACE



**Transaction Medium**  
Green Points



**Value Source**  
Environmental actions (Recycle, Clean up, Green activities, etc.)



**Purpose**  
Promote sustainability & resource reuse



**User Behavior**  
Act → Earn → Exchange → Reuse



**Incentive Mechanism**  
Green Points reward for positive actions



**Environmental Impact**  
High positive impact on campus & environment



**AI Assistance**  
AI Scan, AI Advice, Smart Recommendations



**Resource Flow**  
Green action → Reuse → Reduce waste → Circular economy



**Core Value**  
Environmental + Social Value



### ECOCAMPUS FLOW



Do something green



Earn Green Points



Exchange in Marketplace



Item gets a second life



Create bigger environmental impact



BUILD A GREENER  
CAMPUS TOGETHER



REDUCE WASTE  
IN CAMPUS



ENCOURAGE POSITIVE  
ENVIRONMENTAL ACTIONS



STRENGTHEN CAMPUS  
COMMUNITY



CREATE A SUSTAINABLE  
FUTURE



## WHY WE STAND OUT: SMART & GREEN

### Dual-Core Value Proposition

#### Where is Smart?

- Automated visual verification of eco-actions.
- AI-driven resource routing and price recommendation.
- Real-time supply and demand optimization.



Real-time supply and demand optimization.



#### Where is Green?

- Maximizes campus waste sorting and material recovery.
- Extends product lifecycles and reduces unnecessary consumption.
- Quantifies campus-wide carbon footprint reduction.



Quantifies campus-wide carbon footprint reduction.



**DON'T JUST THROW IT AWAY.  
GIVE IT ANOTHER LIFE.**

**ECOCAMPUS**

Smart Campus · Circular Value · Greener Future

**THANK YOU.**